Lab 04: Single-Area OSPFv2 Configuration & Verification

CCNA Lab Portfolio | Cisco Packet Tracer Documentation

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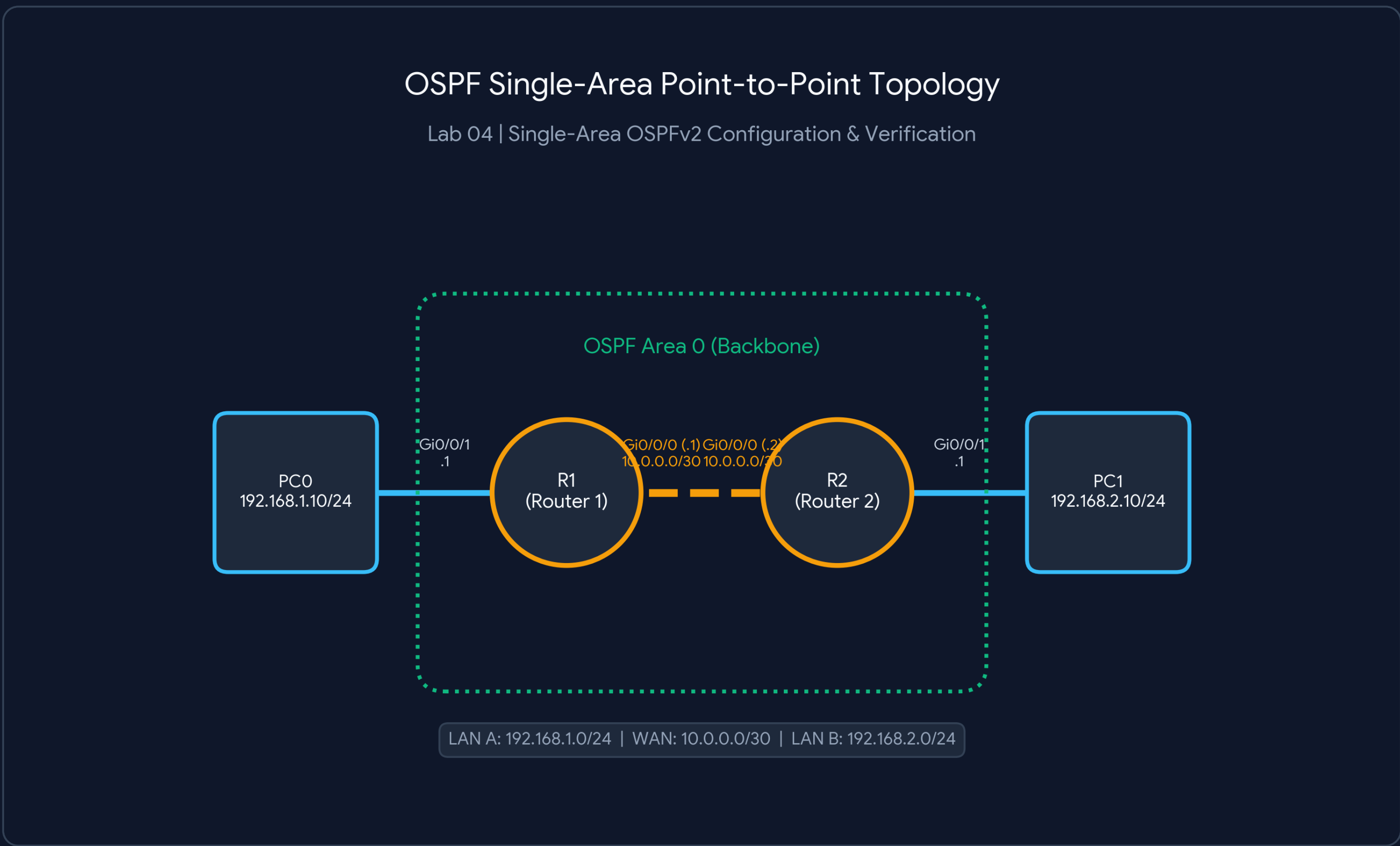
Lab ID / Topic: Lab 04 - Dynamic Routing (OSPFv2)	**Environment:** Cisco Packet Tracer v8.x

1. Executive Summary & Topology Overview

This lab demonstrates the end-to-end implementation of Single-Area Open Shortest Path First version 2 (OSPFv2) across a point-to-point router connection using Area 0 (Backbone Area). The lab covers IPv4 subnetting, wildcard mask calculations, OSPF Router ID selection, inter-router neighbor adjacency establishment, and route verification.



2. IP Addressing & Interface Specification

Device	Interface	IP Address	Subnet Mask	Wildcard Mask	OSPF Zone
R1	Gi0/0/0	10.0.0.1	255.255.255.252 (/30)	0.0.0.3	Area 0
R1	Gi0/0/1	192.168.1.1	255.255.255.0 (/24)	0.0.0.255	Area 0
R2	Gi0/0/0	10.0.0.2	255.255.255.252 (/30)	0.0.0.3	Area 0

Device	Interface	IP Address	Subnet Mask	Wildcard Mask	OSPF Zone
R2	Gi0/0/1	192.168.2.1	255.255.255.0 (/24)	0.0.0.255	Area 0
PC0	NIC	192.168.1.10	255.255.255.0 (/24)	N/A	GW: 192.168.1.1
PC1	NIC	192.168.2.10	255.255.255.0 (/24)	N/A	GW: 192.168.2.1

3. Important Terminologies & Formulas

Wildcard Mask Formula:
Wildcard Mask = 255.255.255.255 - Subnet Mask

Example (/30 Subnet):
255.255.255.255 - 255.255.255.252 = 0.0.0.3

Example (/24 Subnet):
255.255.255.255 - 255.255.255.0 = 0.0.0.255

- **OSPF Area 0:** The backbone area to which all other areas in an OSPF domain must connect.
- **Router ID (RID):** A unique 32-bit identifier for each OSPF router. Priority: Manual Configuration > Highest Active Loopback IP > Highest Active Physical IP.
- **Wildcard Mask:** Inverted mask where '0' means exact match and '255' means ignore/wildcard.

4. Step-by-Step CLI Configuration

Router 1 (R1) Configuration

```
R1# configure terminal
R1(config)# interface GigabitEthernet0/0/0
R1(config-if)# ip address 10.0.0.1 255.255.255.252
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# interface GigabitEthernet0/0/1
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# exit

R1(config)# router ospf 1
R1(config-router)# router-id 1.1.1.1
R1(config-router)# network 10.0.0.0 0.0.0.3 area 0
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# passive-interface GigabitEthernet0/0/1
R1(config-router)# end
R1# write memory
```

Router 2 (R2) Configuration

```
R2# configure terminal
R2(config)# interface GigabitEthernet0/0/0
```

```
R2(config-if)# ip address 10.0.0.2 255.255.255.252
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# interface GigabitEthernet0/0/1
R2(config-if)# ip address 192.168.2.1 255.255.255.0
R2(config-if)# no shutdown
R2(config-if)# exit

R2(config)# router ospf 1
R2(config-router)# router-id 2.2.2.2
R2(config-router)# network 10.0.0.0 0.0.0.3 area 0
R2(config-router)# network 192.168.2.0 0.0.0.255 area 0
R2(config-router)# passive-interface GigabitEthernet0/0/1
R2(config-router)# end
R2# write memory
```

5. Verification & Troubleshooting Commands

```
R1# show ip ospf neighbor
Neighbor ID     Pri   State           Dead Time   Address         Interface
2.2.2.2           0   FULL/ -         00:00:34    10.0.0.2        GigabitEthernet0/0/0

R1# show ip route ospf
O    192.168.2.0/24 [110/2] via 10.0.0.2, 00:05:12, GigabitEthernet0/0/0

C:\> ping 192.168.2.10 (From PC0)
Pinging 192.168.2.10 with 32 bytes of data:
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126
Packet sent: 4, Received: 4, Lost: 0 (0% loss)
```